Thanh Dat Le

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Summary

New Grad (M.S. Computer Science, Northeastern University, Expected 2027 | Dean's List) with industry experience in Computer Vision and AI engineering at HQSoft and AI research through the AIMA–CMU mentorship program. Builds end-to-end ML systems spanning LLM fine-tuning, agentic RAG pipelines, VLM evaluation, and MLOps deployments on AWS.

Education

Northeastern University Boston, MA
B.S. & M.S. in Computer Science (Combined Program) *2024 – Expected 2027*

- **Honors:** Dean's List 2025–2026 — Recipient of competitive merit scholarship, Northeastern University
- **Coursework:** Machine Learning, Deep Learning, Reinforcement Learning, Computer Vision, Algorithms

Technical Skills

AI / ML: PyTorch, LLM Fine-Tuning (LoRA/Unsloth), Agentic RAG, LangGraph, FAISS, Qdrant, VLMs, YOLOv11, OpenCV, Transformers
Languages: Python, SQL, Java, C/C++
MLOps / Cloud: Docker, AWS (ECR, S3), GitHub Actions (CI/CD), DVC, FastAPI, Google Cloud
Data: PostgreSQL (pgvector), MySQL, Vector Embeddings, BM25

Experience

F&B Nexus Ho Chi Minh City, Vietnam
Co-founder — Multi-Tenant Restaurant Operations Platform *Oct 2026 – Present*
Java 21 · Spring Boot 3.3 · PostgreSQL · RabbitMQ · Docker Compose · Google Cloud · CI/CD

- Architected and deployed a multi-service platform serving **50+ real users**; containerized all services with **Docker Compose** and migrated production infrastructure from AWS EC2 to **Google Cloud Compute Engine**, tuning per-service memory limits after OOM failures.
- Built a reliable async event pipeline using a **transactional outbox** pattern with **RabbitMQ**; managed all schema changes through a forward-only **Liquibase** changelog (expand–migrate–contract) to ensure zero-downtime deployments.

HQSoft Ho Chi Minh City, Vietnam
AI Engineer Intern — Distribution Management System (DMS) *Jun 2026 – Sep 2026*

- Retail shelf audits were manual and error-prone; trained two **YOLOv11** models for product-level **object detection** and **shelf row segmentation**, automating planogram compliance verification from mobile field photos.
- Deployed models were too large for field devices; applied **pruning and quantization** to reduce model size and improve inference speed, achieving **mAP@50 of 80–90%** on resource-constrained hardware.
- Mobile cameras captured only partial shelf views, blocking full-shelf analysis; built an **OpenCV** stitching pipeline using **SIFT** feature matching and **homography** estimation to reconstruct complete shelf **panoramas** from overlapping captures.
- Manual shelf measurement was time-consuming and error-prone; developed a **reference-object calibration** system using known-size products as anchors to estimate real-world shelf dimensions directly from images.
- Integrated all modules into an **end-to-end planogram verification workflow** served via **FastAPI**, enabling real-time compliance checks within the DMS platform for field sales reps.

AIMA Research Mentorship Program Remote
AI Research Mentee — AIMA × Carnegie Mellon University *Jan 2026 – Present*

- VLM robustness under real-world degradation was largely unexplored; engineered a configurable noise pipeline (Gaussian, blur, JPEG, contrast) to benchmark **Ferret-7B** and **NVIDIA DAM 3B** on region-description quality across corruption levels.
- Five benchmark datasets (**LVIS, PACO, COCO, DLC-Bench, Ref-L4**) used incompatible formats; engineered a **UniversalDataset** abstraction supporting pickle and COCO JSON, reducing per-dataset integration from hours to minutes.
- Standard metrics failed to capture semantic degradation; implemented **LLM-as-judge** via **CLAIR** (Llama 3.1 8B) alongside BLEU-4, METEOR, ROUGE-L, CIDEr; deployed Ferret-7B with **4-bit quantization** and **Meta SAM** within a 12-hour Kaggle P100 session.
- Co-authored a research manuscript on VLM robustness under visual and spatial corruptions submitted to **IEEE Access**.

Projects

Adaptive Agentic RAG — Clinical Diagnosis System GitHub 2025
LangGraph · Qdrant · BM25 · LoRA/Unsloth · Google AI Studio · DVC · AWS S3

- Off-the-shelf LLMs failed on vague clinical queries; fine-tuned 4 models (**Gemma 2 9B, Llama 3.1 8B, Qwen3 8B, Qwen2.5 7B**) via **LoRA/Unsloth** and designed a **3-route adaptive LangGraph pipeline** routing vague queries through an **agentic enricher** that generates clinical assumptions grounded with live web search before retrieval.
- Single-modal retrieval produced low-relevance results; built a **hybrid retriever** fusing **BM25** and **Qdrant dense search** with **Reciprocal Rank Fusion**, feeding graded documents to **Gemma 4 26B** via **Google AI Studio API** for generation.
- Benchmarked on a **279-query stratified set**: **79.2% disease hit rate** vs. **33.3% baseline (+45.9 pp)** and **100% route accuracy** (24/24); versioned all artifacts via **DVC / AWS S3**.

Animals Detection MLOps Pipeline GitHub 2026
PyTorch · FastAPI · Docker · AWS ECR · DVC · GitHub Actions

- Notebook-based CV workflows lacked reproducibility and deployment readiness; built a **PyTorch** Transfer Learning classifier orchestrated by **DVC**, automated train–build–push via **GitHub Actions CI/CD** to **AWS ECR**, and deployed a **FastAPI** server with **OOD detection** to flag low-confidence predictions.